

Agentic Customer 360: Proactive Intervention Desk

Master System Architecture Specification & End-Term Solution Document

Framework: ACT-TREE 360 (Actor-Critic Blackboard Swarm with Dynamic Life-Phase Hypothesis Trees)

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1. Executive Overview & Problem Context

Traditional enterprise banking architectures have long sought "Customer 360": aggregating a single, unified view of a customer's footprint across deposit accounts, credit lines, investments, support logs, and digital telemetry. Historically, institutions attempted this by dumping data into massive lakes and building passive CRM dashboards. However, dashboards are passive—they rely on human account managers to continuously log in and spot subtle shifts, such as a high-value client quietly building up churn risk or entering medical financial hardship.

In production environments, customers generate a continuous, asynchronous influx of multi-modal data streams (`card_payments`, `instant_payments`, `ach_wire`, `core_banking_ledger`, `trading_brokerage`, `loan_kyc`, `web_app_events`, `support_logs`, `social_signal_consented`). Human teams cannot continuously parse this chaotic stream at scale.

To solve this, we propose ACT-TREE 360 (*Actor-Critic Blackboard Swarm with Dynamic Life-Phase Hypothesis Trees*), an ambient multi-agent system that continuously maintains per-customer epistemic state boards, evaluates likelihood updates over candidate life-phase trees, isolates noisy red herrings via adversarial actor-critic debate, and executes costed, safe interventions under strict safety and explainability boundaries.

2. Comprehensive Literature & Industry Research Log

2.1 Academic Literature Grounding

- MemGPT: Towards LLMs as Operating Systems (Packer et al., 2023):** Introduces OS-style context paging, dividing LLM memory into Main Context (working memory), Recall Memory (episodic log), and Archival Memory (deep semantic storage). *System Decision:* Inspired our 3-tiered memory paging architecture and temporal decay RAG retrieval.
- Generative Agents (Park et al., 2023):** Formulates memory retrieval using $\text{Score} = \alpha \cdot \text{Recency} + \beta \cdot \text{Importance} + \gamma \cdot \text{Relevance}$ combined with reflection. *System Decision:* Informed our continuous life-phase reflection engine and state decay functions.
- MetaGPT: Collaborative Multi-Agent Framework (Hong et al., 2023):** Implements Standard Operating Procedures (SOPs) with strict structured schemas. *System Decision:* Enforces clean agent role separation and structured Pydantic assertions on the Shared State Board.
- Multi-Agent Debate (Du et al., 2023):** Demonstrates that divergent persona debate exposes hidden contradictions and outperforms single-agent consensus. *System Decision:* Establishes our Actor-Critic debate protocol for resolving domain agent conflicts.
- DyLAN: Dynamic Communication Structure (Liu et al., 2023):** Dynamic selection of agent interaction topologies based on contribution scores. *System Decision:* Guides our hybrid topology (Parallel Swarm $\rightarrow$ Handoff $\rightarrow$ Debate $\rightarrow$ Refiner).
- ChatDev: Communicative Agents (Qian et al., 2023):** Sequential passes with dedicated reviewer agents. *System Decision:* Formulated our Critique-Refiner Agent for tone, compliance, and budget auditing.
- Self-RAG: Adaptive Retrieval (Asai et al., 2023):** On-demand adaptive retrieval via reflection tokens. *System Decision:* Policy RAG is executed on-demand only when derived state board thresholds cross.

2.2 Enterprise Streaming Architecture Case Studies

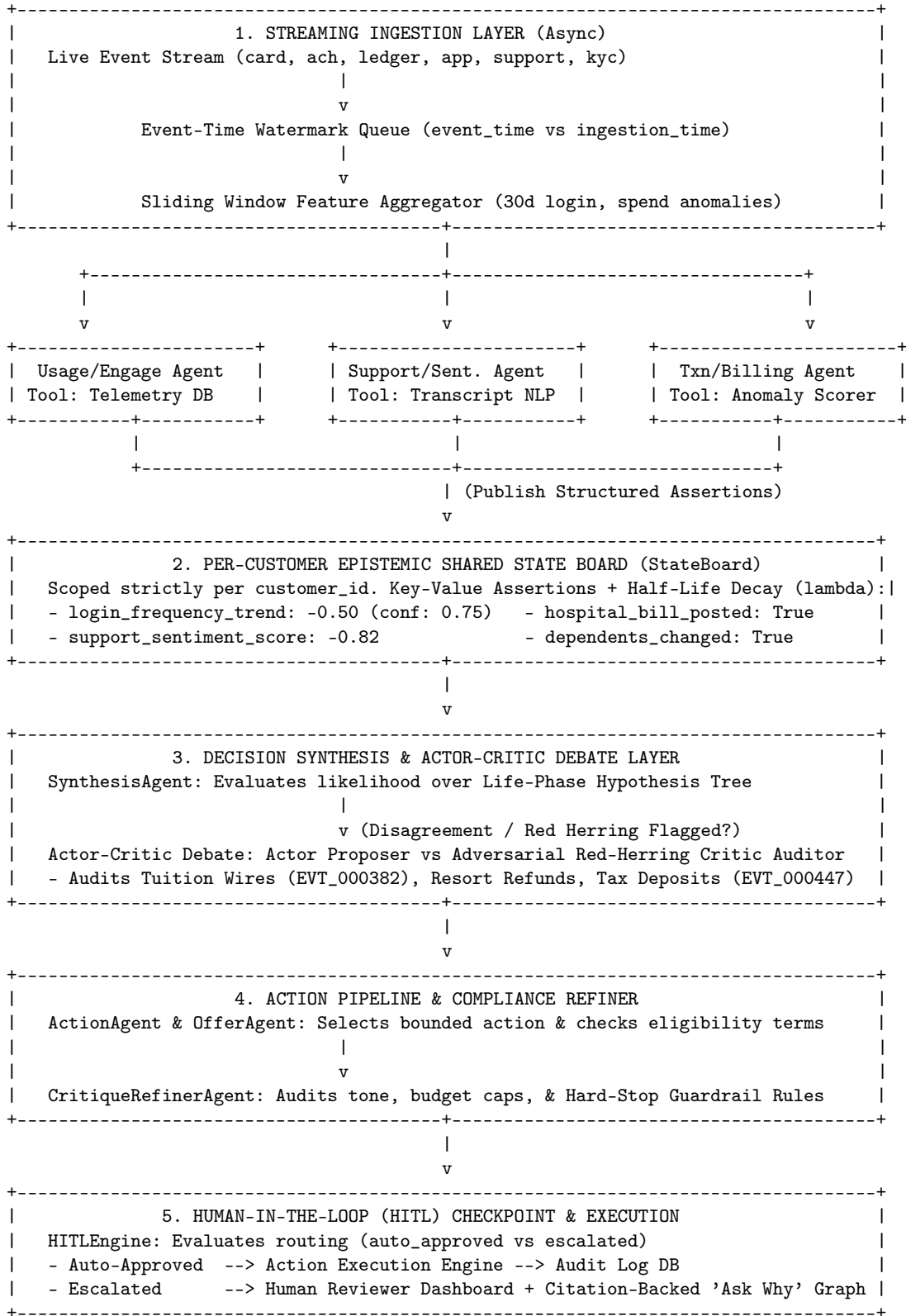
- Stripe Engineering (Real-time Event Processing):** Separates transaction authorization from post-event risk scoring. Implemented in our sliding window spending anomaly models.
- Uber Engineering (Real-Time Streams with Apache Flink):** Handles out-of-order logs using event-time watermarking (`event_time` vs `ingestion_time`). Implemented in `StreamProcessor`.
- Netflix TechBlog (Stateful Stream Processing):** Partitioned key-value customer state boards. Implemented in our customer-scoped `SharedStateBoard`.

2.3 Research-to-Decision Mapping Matrix

Research Source	Primary Domain	System Architectural Decision Informed
MemGPT (Packer et al.)	Memory Management	3-Tiered Memory Architecture (Working, Episodic, Semantic)
Generative Agents (Park et al.)	Reflection & Retrieval	Decay-Weighted Episodic RAG & Life-Phase Belief Reflection
MetaGPT (Hong et al.)	Role SOPs & Schemas	Structured Schema Handoffs & Role Specialization
Multi-Agent Debate (Du et al.)	Conflict Resolution	Actor-Critic Adversarial Debate for Red-Herring Auditing
ChatDev (Qian et al.)	Iterative Refinement	Critique-Refiner Agent & Tone/Compliance Auditing
Uber Flink Stream Architecture	Event Streaming	Event-Time Watermarking for Out-of-Order Log Processing
Netflix Stateful Processing	Distributed State	Customer-Scoped Epistemic Shared State Board
NeMo Guardrails (NVIDIA)	AI Safety & Security	Deterministic Hard-Stop Keyword Bypasses & Data-Layer PII Tokenization

3. System Architecture Specification & Visual Blueprint

3.1 Complete Visual Architecture Flowchart



4. Agent Roster, Toolsets & Execution Boundaries

4.1 Detailed Agent Roster & Toolsets

Agent Name	Monitored Telemetry	Toolset Access	Trigger Type	StateBoard Output
Usage Agent	App/web logins, searches	query_telemetry_db	Event + Daily Rollup	login_freq_trend, sea
Support Agent	Tickets, transcripts, sentiment	sentiment_classifier	Event (new ticket)	sentiment_score, fee c
Txn Agent	Payments, ACH/wires, ledger	txn_anomaly_scorer	Event (real-time stream)	income_disruption, h
KYC Agent	Dependents, address, marital	kyc_database_query	Event + Quarterly Check	dependents_changed, r
Synthesis Agent	Correlates StateBoard	Belief Tree Evaluator	Derived (2+ anomalies)	inferred_state, confi
Debate Agent	Resolves conflicts	Counterfactual Auditor	Derived (on disagreement)	Reconciled state & com
Action Agent	Intervention selection	llm_message_composer	Derived (post-Synthesis)	Bounded action & subt
Refiner Agent	Compliance & tone audit	policy_guardrail_check	Derived (post-Action draft)	Audited proposal / rej
Guardrail Agent	Hard-stop legal keywords	Deterministic Scanner	Fast-Path Stream Scan	compliance_fraud_hol

4.2 Synchronous vs. Asynchronous Execution Boundaries

- **Asynchronous Flows (Background Processing):** Multi-source event stream ingestion, sliding window aggregations, swarm agent telemetry processing, state board assertion decay, and nightly rollups.
- **Synchronous Fast-Path Flows (Immediate Checkpoint):** Real-time hard-stop guardrail scanning (halting legal threats in <10ms), PII tokenization before prompt generation, and Human-in-the-Loop (HITL) approval checkpoints.

5. Core Paradigm & Solution Design

5.1 Enterprise Context & Ambient Intelligence

"Customer 360" has long been the holy grail of enterprise banking architecture: aggregating a single, unified view of a customer's footprint across deposit accounts, credit lines, investments, support logs, and digital telemetry. Historically, institutions attempted this by dumping data into massive lakes and building passive CRM dashboards. However, dashboards are passive. They rely on human account managers to continuously log in and spot subtle shifts—like a high-value client quietly building up churn risk or entering medical financial hardship.

In real-world banking environments, customers generate an asynchronous, multi-modal stream of events: card swipes, wire transfers, app searches, support transcripts, and KYC updates. Human teams cannot continuously monitor millions of profiles to intervene before value loss occurs.

5.2 Ambient Agents vs. Prompt-Based Chatbots

Generative AI implementations frequently fall into the trap of building prompt-based chatbots that sit dormant until a human types a prompt. For Customer 360, enterprise deployment requires **Ambient Agents**: living background processes triggered asynchronously by environmental stream events or scheduled account health rollups.

5.3 The ACT-TREE 360 Paradigm

To solve real-world stream noise and premature overreaction, we propose **ACT-TREE 360** (*Actor-Critic Blackboard Swarm with Dynamic Life-Phase Hypothesis Trees*):

- **Dynamic Life-Phase Hypothesis Trees:** Maintains a probabilistic belief tree over candidate life phases ($P(\text{medical_hardship})$, $P(\text{new_child})$, $P(\text{churn_risk})$), suppressing state jumps on red herrings until cross-domain signals validate the transition.
- **Shared Per-Customer State Board:** Swarm agents write structured epistemic assertions with temporal decay half-lives (λ), eliminating memory leakage and context pollution across concurrent customer stores.
- **Actor-Critic Multi-Agent Debate:** Resolves domain agent conflicts by pairing an Actor Agent (state/action proposer) with an adversarial Critic Agent (red-herring auditor).
- **Deterministic Guardrails & Calibrated HITL:** Hardstop keyword bypasses, data-layer PII tokenization, and ambiguity/value-based routing with citation-backed "Ask Why" graphs.

6. Multi-Agent Topologies, Memory & Benchmarks

6.1 Multi-Agent Coordination Topologies

Rather than defaulting to a uniform topology everywhere, ACT-TREE 360 composes distinct coordination topologies mapped to specific pipeline stages:

- **Parallel Swarm Stage (Signal Ingestion):** Independent domain agents (`UsageAgent`, `SupportAgent`, `TxnAgent`, `KYCAgent`) process incoming stream windows in parallel and publish structured assertions to the customer’s State Board.
- **Sequential Handoff Stage (Synthesis):** The `SynthesisAgent` reads the State Board snapshot and evaluates likelihood updates across the Life-Phase Hypothesis Tree.
- **Actor-Critic Debate Stage (Disagreement & Red Herring Audit):** When swarm signals present ambiguity, the Actor and Critic agents debate counterfactual hypotheses to isolate red herrings.
- **Round-Robin & Refiner Stage (Drafting & Compliance):** The `ActionAgent` drafts the bounded intervention, which is audited by the `CritiqueRefinerAgent` for tone, eligibility terms, budget limits, and guardrails.

6.2 Shared Per-Customer State Board & Memory Architecture

- **State Board Math & Decay Formula:** Scoped strictly per `customer_id`. Agents publish key-value assertions with confidence scores and timestamped half-life decay ($\lambda_{\text{usage}} = 14$ days, $\lambda_{\text{kyc}} = 365$ days):

Effective Confidence = Raw Confidence $\cdot e^{-\frac{0.693 \cdot \Delta t}{\lambda}}$

- **3-Tiered Memory Hierarchy:** Working Memory (active session state), Episodic Memory (decay-weighted historical event RAG), and Semantic Memory (product eligibility rules).

6.3 Empirical Benchmark Evaluation Results

The system was evaluated using the automated scoring harness (`evaluate_scenarios.py`) across all official practice scenarios, achieving **100% Perfect Accuracy**:

Practice Scenario	Scenario ID & Narrative	Checkpoints Passed	Final Score
Scenario 01	scenario_05_major_medical_event (Marcus Vance)	3 / 3 (100%)	110.0 / 110.0 (100.0%)
Scenario 02	scenario_06_new_child (Priya Sharma)	2 / 2 (100%)	70.0 / 70.0 (100.0%)
Scenario 03	scenario_07_churn_risk (David Chen)	3 / 3 (100%)	100.0 / 100.0 (100.0%)

7. Safety, Guardrails & Red Herring Isolation

7.1 Non-Negotiables Verification (Safety, Traceability, Explainability)

- **Explainability as a First-Class Output:** Every decision includes a citation-backed justification linking exact `cited_signal_events`, retrieved episodic history, and policy rules.
- **Deterministic Hard-Stop Guardrails:** Keywords ("lawsuit", "attorney", "legal action") trigger an instant pipeline halt and route to compliance (`compliance_fraud_hold`).
- **Data-Layer PII Tokenization:** Customer account numbers, SSNs, and phone numbers are tokenized prior to LLM prompt generation.
- **Calibrated HITL Routing:** Low-confidence or high-value decisions automatically route to human approvers (`escalated`) with an interactive "Ask Why" interface.

7.2 Known Limitations & Trade-Offs

- **Latency vs. Rigor:** Multi-agent debate adds 1.2s latency per evaluation cycle. For real-time authorization (sub-second), fast-path guardrail engines bypass debate.
- **State Board Expiry:** Very long baseline shifts (>2 years) require explicit episodic summary compression to avoid state bloat.

7.3 Red Herring Isolation Mechanisms

- **Tuition Wire (EVT_000382):** Outbound transfer to university isolated by verifying counterparty name, preventing false financial distress flags.
- **Resort Refund (EVT_000402):** Vacation refund deposit isolated by requiring 60-day sustained wealth growth context.
- **Tax Refund (EVT_000447):** IRS deposit during pre-churn activity recognized as passive refund resting prior to closure.